Riparian habitat restoration increases the availability and occupancy of Yellow-breasted Chat territories but brood parasitism is the primary influence on reproductive performance

Article *in* The Condor · August 2020

DOI: 10.1093/condor/duaa038

THE CONDOR
Ornithological Applications
AmericanOrnithology.org

Volume 122, 2020, pp. 1–16
DOI: 10.1093/condor/duaa038

RESEARCH ARTICLE

Riparian habitat restoration increases the availability and occupancy of Yellow-breasted Chat territories but brood parasitism is the primary influence on reproductive performance

Timothy R. Forrester,[1,a,*] David J. Green,[1] René McKibbin,[2] A. Michael Bezener,[3] and Christine A. Bishop[4,*]

[1] Centre for Wildlife Ecology, Department of Biological Sciences, Simon Fraser University, Burnaby, British Columbia, Canada
[2] Canadian Wildlife Service, Environment and Climate Change Canada, Delta, British Columbia, Canada
[3] ECOmmunity Place, En'owkin Centre, Penticton, British Columbia, Canada
[4] Wildlife Research Division, Science and Technology Branch, Environment and Climate Change Canada, Delta, British Columbia, Canada
[a] Current address: Montana Cooperative Wildlife Research Unit, University of Montana, Missoula, Montana, USA
*Corresponding authors: Timothy R. Forrester, timothy.forrester@umontana.edu; Christine A. Bishop, christine.bishop@canada.ca

Submission Date: January 23, 2020; Editorial Acceptance Date: June 8, 2020; Published August 21, 2020

ABSTRACT

Implementation and evaluation of conservation efforts requires an understanding of the habitat selection and reproductive success of endangered populations. As populations recover, established territory holders may force new arrivals into lower quality habitat, which can reduce reproductive success, especially in disturbed landscapes where suitable habitat is scarce. The endangered Western Yellow-breasted Chat (*Icteria virens auricollis*) population in the fragmented riparian zone of the Okanagan Valley, British Columbia, Canada, has rapidly increased in response to habitat restoration. During this population increase from 2002 to 2014, we monitored 485 chat nests in 157 breeding territories to evaluate the influences of habitat selection, habitat restoration, and brood parasitism by Brown-headed Cowbirds (*Molothrus ater*) on reproductive performance. We found that, in protected reference sites, breeding territories that were occupied in the early years of the study had higher percent shrub cover than territories that were first occupied in the later years of the study, indicating that chats preferred territories with high shrub cover. Conversely, in restoration sites, later-occupied territories had similarly high shrub cover as earlier-occupied territories, suggesting that restoration activities enabled chats to continually settle in territories with high shrub cover. Yet, we did not find strong evidence that nest site vegetation characteristics or habitat restoration influenced reproductive performance. Instead, the high rate of brood parasitism (49%), which reduced nest success and productivity, was the dominant influence on reproductive performance. However, this recovering population still had high daily nest survival (0.974) and productivity (2.72 fledglings per successful nest) compared with other riparian songbirds and the high parasitism rate did not prevent the population from increasing. Thus, conservation efforts for Yellow-breasted Chats should focus on restoring riparian shrubs, even within heavily developed landscapes, to increase the number of potential breeding territories, while also quantifying how brood parasitism influences reproductive performance.

Keywords: Brown-headed Cowbird, daily nest survival, fledglings, habitat quality, habitat selection, population response, riparian songbird, shrub cover

LAY SUMMARY

- In landscapes where undisturbed, high-quality habitat is scarce, some birds may be forced to occupy low-quality habitat. Habitat restoration may provide more high-quality habitat.
- The endangered population of Yellow-breasted Chats in British Columbia increased in abundance from 2002 to 2014 in response to habitat restoration efforts. We investigated 2 questions: (1) Do chats prefer to nest in areas with certain vegetation characteristics? and (2) What effects do vegetation characteristics and parasitism by Brown-headed Cowbirds, which lay their eggs in chat nests, have on chat reproductive success?
- Settlement patterns indicated that chats prefer to occupy breeding territories with high shrub cover. However, Brown-headed Cowbirds parasitized half of all chat nests, and whether or not a nest was parasitized was more important than vegetation characteristics in determining how many chat fledglings were produced.
- Habitat restoration efforts provided more suitable habitat for this recovering population but did not reduce how often chat nests were parasitized. Chats in this population still have relatively high reproductive success and therefore are not as vulnerable to cowbird parasitism as other songbirds.

La restauración de hábitat ribereño aumenta la disponibilidad y la ocupación de los territorios de *Icteria virens auricollis*, pero el parasitismo de nidada es la principal influencia en el desempeño reproductivo

RESUMEN

La implementación y evaluación de los esfuerzos de conservación requiere un entendimiento de la selección de hábitat y del éxito reproductivo de las poblaciones en peligro. A medida que las poblaciones se recuperan, los poseedores de territorios establecidos pueden forzar a los que recién llegan hacia hábitat de menor calidad, lo que puede reducir el éxito reproductivo, especialmente en paisajes disturbados donde el hábitat adecuado es escaso. La población en peligro de *Icteria virens auricollis* de la zona ribereña fragmentada del Valle de Okanagan, Columbia Británica, Canadá, ha aumentado rápidamente en respuesta a la restauración de hábitat. Durante este aumento poblacional, desde 2002–2014, monitoreamos 485 nidos de *I. v. auricollis* en 157 territorios reproductivos para evaluar las influencias de la selección de hábitat, la restauración de hábitat y el parasitismo de nidada por *Molothus ater* sobre el desempeño reproductivo. Encontramos que, en sitios protegidos de referencia, los territorios reproductivos que fueron ocupados durante los primeros años del estudio tuvieron un porcentaje de cobertura de arbustos más alto que los territorios que fueron ocupados por primera vez en los años posteriores del estudio, indicando que los individuos de *I. v. auricollis* prefirieron los territorios con alta cobertura de arbustos. Por el contrario, en los sitios restaurados, los territorios ocupados más tarde tuvieron una cobertura de arbustos igualmente alta que los territorios ocupados al inicio, sugiriendo que las actividades de restauración les permitieron a los individuos de *I. v. auricollis* establecerse de modo continuo en territorios con alta cobertura de arbustos. Aun así, no encontramos evidencia fuerte de que las características de la vegetación del sitio de anidación o que la restauración del hábitat influenciaron el desempeño reproductivo. En cambio, la alta tasa de parasitismo de nidada (49%), la que redujo el éxito y la productividad del nido, fue la influencia dominante en el desempeño reproductivo. Sin embargo, esta población en recuperación todavía tenía una alta supervivencia (0.974) y productividad (2.72 volantones por nido exitoso) diaria del nido, en comparación con otras aves canoras ribereñas, y la alta tasa de parasitismo no evitó que la población aumentara. Por ende, los esfuerzos de conservación para *I. v. auricollis* deberían enfocarse en restaurar los arbustos ribereños, incluso dentro de paisajes muy desarrollados, para aumentar el número de territorios reproductivos potenciales, mientras que también deberían cuantificar cómo el parasitismo de nidada influencia el desempeño reproductivo.

Palabras clave: ave canora ribereña, calidad de hábitat, cobertura de arbustos, *Molothus ater*, respuesta poblacional, selección de hábitat, supervivencia diaria del nido, volantones

INTRODUCTION

Effective conservation planning and evaluation of habitat restoration require long-term monitoring of the habitat selection and reproductive success of endangered populations (Johnson 2007, Peak and Thompson III 2014). Habitat selection is a hierarchical process involving innate and learned decisions that animals use to choose habitat components at different spatial scales, while habitat quality describes the degree to which the habitat provides conditions appropriate for individuals and populations to reproduce and persist (Orians and Wittenberger 1991, Hall et al. 1997). Differences in habitat quality can lead to differences in reproductive success (Martin 1987, Andrén 1990, Donovan et al. 1995), so natural selection should favor traits linked to the ability to recognize high-quality habitat (Fretwell and Lucas 1969, Martin 1998, Clark and Shutler 1999). However, the consequences of habitat selection for reproductive performance in disturbed and developed areas, as well as in areas with high abundances of predators or parasites, are unclear.

The habitat features that influence settlement are often used to guide conservation efforts for animal populations (Powell and Steidl 2000, Davis et al. 2014, White et al. 2019). The presence or abundance of a species in a particular site is often presumed to mean that the habitat is of high quality. Abundance can sometimes be an indicator of habitat quality (Bock and Jones 2004), but may not be when animals are forced to occupy disturbed and/or low-quality habitat (Van Horne 1983). Among territorial species, breeding territories that are occupied first are usually considered to be preferred and low-quality territories may only be occupied when density is high (Lanyon and Thompson 1986, Currie et al. 2000, Joos et al. 2014). The ideal despotic distribution theory predicts that established territory holders will exclude newly arriving individuals from the highest quality breeding territories, so individuals occupying suboptimal territories may have lower reproductive performance (Fretwell and Lucas 1969, Ferrer and Donazar 1996). Preferred breeding territories are sometimes associated with higher reproductive success (Högstedt 1980, Martin 1998, Tschumi et al. 2014), but often are not (Chalfoun and Schmidt 2012, Germain and Arcese 2014). Habitat features may not influence breeding success when habitat disturbances result in species being unable to assess the true quality of the habitat (Misenhelter and Rotenberry 2000, Battin 2004). Thus, assessment of the fitness consequences of habitat selection is critical for guiding management of endangered populations, especially in heavily disturbed areas.

Conservation efforts that use habitat restoration techniques to create new habitat or improve existing habitat can increase the density and abundance of songbirds (Krueper et al. 2003, Gardali et al. 2006, Earnst et al. 2012). Forrester et al. (2017) previously documented the drastic increase in abundance (from near local extirpation) of the neotropical migrant Western Yellow-breasted Chat (*Icteria virens auricollis*; hereafter "chats") population in remnant riparian habitat of the south Okanagan Valley of British Columbia, Canada. Abundance increased rapidly in restoration sites where the grazing of livestock was limited, but not in protected reference sites (Environment and Climate Change Canada 2016, Forrester et al. 2017). This population has high annual return rates and individuals often show high fidelity to breeding territories (McKibbin and Bishop 2008, 2012a), suggesting that specific habitat features are important to chat territory occupancy. This population also has a high rate of brood parasitism by Brown-headed Cowbirds (*Molothrus ater*; hereafter "cowbirds"; Forrester et al. 2017). Here, we used the unique opportunity offered by the rapid increase in abundance that occurred during our study to evaluate the influences of habitat selection, habitat restoration, and brood parasitism on the reproductive performance of chats in this recovering population.

We address 2 main questions. First, as chat abundance increased, did chats preferentially settle in breeding territories with certain vegetation characteristics? Given the high territory fidelity in this population, we predicted that habitat characteristics influencing chat breeding territory selection (e.g., percent shrub cover) would be higher in territories that were settled earlier in the study (Figure 1). However, we also predicted that this effect would be stronger in reference sites than in restoration sites due to regrowth of riparian vegetation in the restoration sites and therefore creation of more suitable breeding areas over the study period (Figure 1). Second, what were the effects of nest site vegetation characteristics, breeding territory selection, habitat restoration, and brood parasitism on the reproductive performance of chats as their abundance increased? The heavily fragmented and developed landscape of the Okanagan Valley increased the likelihood of reduced reproductive performance at higher population density if chats were forced to occupy low-quality habitat (Bock and Jones 2004). We predicted that chats breeding in territories that were occupied earlier in the study would have higher reproductive performance than chats breeding in newly occupied territories, but also that this effect would be stronger in reference sites than in restoration sites where habitat improvement could be expected to result in increased reproductive performance (Figure 1). However, we also predicted that if chats were unable to select territories that would minimize the chance of brood parasitism, then there would be no relationship between habitat selection

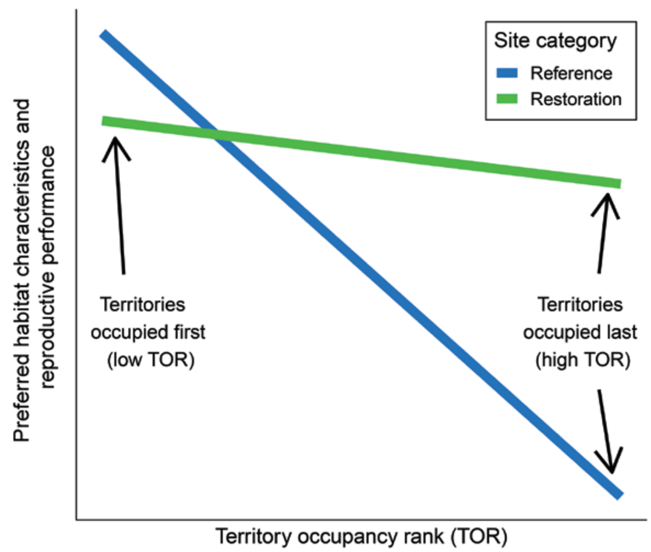

FIGURE 1. Prediction for how preferred habitat characteristics of breeding territories and reproductive performance of Yellow-breasted Chats would vary with TOR (the year a territory was first occupied by a breeding chat) as the population experienced a rapid increase in abundance. We predicted that preferred habitat characteristics and reproductive performance would be higher in territories that were occupied earlier in the study (i.e. those with a lower TOR) when chat abundance was low, as compared with those territories that were not occupied until later in the study (i.e. those with a higher TOR) when chat abundance was much higher. We also predicted this effect would be stronger in reference sites than in restoration sites where habitat restoration was predicted to gradually create more suitable breeding habitat.

and reproductive performance due to the strong influence of parasitism on reproductive performance.

METHODS

Study Population

The endangered population of chats in the south Okanagan Valley of British Columbia, Canada, has been the subject of conservation efforts that have resulted in an increase in abundance from near extirpation (Forrester et al. 2017). Throughout the 1990s there were ≤20 pairs in the province and this small population was listed as endangered in 2001 (Environment and Climate Change Canada 2016). The population increased to ~100 pairs during 2001–2013 as a result of using cattle exclusion as a restoration tool in remnant riparian vegetation (Environment and Climate Change Canada 2016, Forrester et al. 2017). Restoration activities aimed to increase the understory shrub layer of degraded riparian areas by excluding cattle because chats in the Okanagan Valley occupy territories that have greater percent shrub cover (i.e. surface area coverage) and lower percent tree cover than randomly selected plots, indicating a preference for high shrub cover (McKibbin and Bishop 2010). They also place their open-cup nests almost exclusively (99% of nests) in thick shrub patches (McKibbin and Bishop 2010). Chats in this population have high apparent survival (Φ = 0.65) for a temperate breeding neotropical migrant and often show high fidelity to specific study sites and/or breeding territories (McKibbin and Bishop 2012a). For example, 12 of 17 banded males showed fidelity to a specific breeding territory for between 2 and at least 5 yr (McKibbin and Bishop 2012a), and one male bred in the same territory for at least 6 yr (McKibbin and Bishop 2008). This suggests that specific habitat features of breeding territories may be influential to chat territory occupancy and reproductive performance.

Study Area

The Okanagan Valley of south central British Columbia, Canada, is an arid valley that connects the intermontane shrub steppe of the Great Basin to the south with the boreal forests to the north (Cannings et al. 1987). This narrow valley contains a mosaic of ecosystems that provide breeding habitat to >200 species of birds (Cannings et al. 2010), part of one of the most irreplaceable avian communities in the country (Warman et al. 2004). Riparian vegetation in the Okanagan Valley has been destroyed and degraded as a result of river channelization, urbanization, and agricultural development (Lea 2008). Historically, nutrient-rich floodplains supported riparian vegetation that followed the length of the Okanagan River. Riparian forests were dominated by black cottonwood (*Populus trichocarpa*) with a dense understory of wild rose, red osier dogwood (*Cornus sericea*), and willow (*Salix* spp.). Swamp wetlands were dominated by water birch (*Betula occidentalis*; Cannings et al. 1987). However, rapid human settlement and development has led to the loss of 58% of black cottonwood forest and 92% of water birch swamp vegetation communities since the mid-1800s (Lea 2008). The remaining riparian vegetation has been severely fragmented and degraded by urban and agricultural activities, road building, and grazing (Lea 2008). Destruction and fragmentation of riparian vegetation has putatively been a major factor in population declines of some riparian songbird species in the Okanagan Valley, including the endangered chat (Environment and Climate Change Canada 2016).

We conducted our study in 6 riparian sites, ranging in elevation from 280–344 meters above sea level, in the southern part of the Okanagan Valley (Figure 2). The northernmost site was at 49.4878°N, 119.6171°W and the southernmost site was at 49.0730°N, 119.5056°W (Figure 2). These 6 sites represent the majority of chat habitat in the Okanagan Valley except for small fragments of inaccessible private property. We classified our study sites into 3 categories: reference (n = 3), restoration (n = 2), or currently grazed (n = 1). Five of the study sites ranged in size

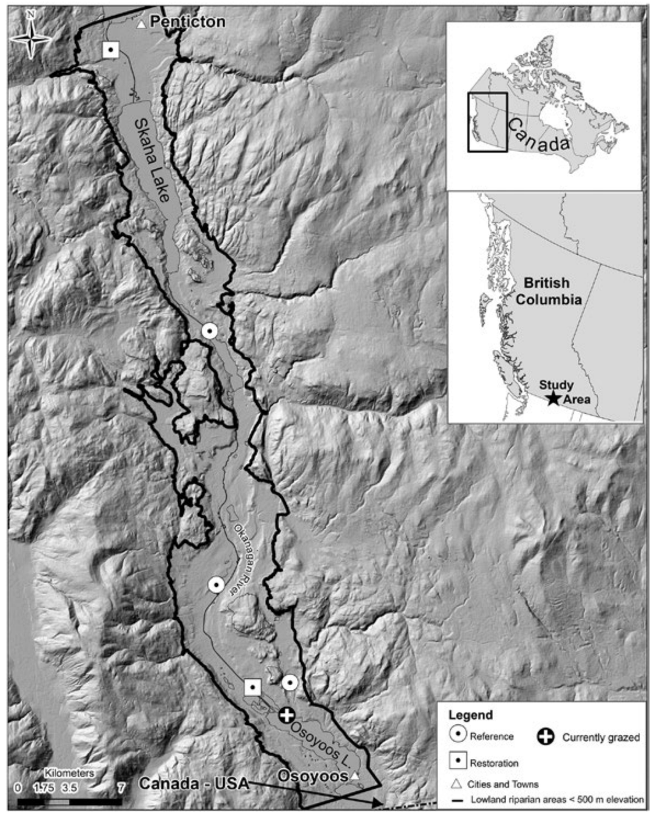

FIGURE 2. Riparian study area showing study sites in the Okanagan Valley, British Columbia, Canada. Reference sites gained protection at varying times after the 1950s as provincial parks or private reserves and were used as a reference of the natural state of the fragmented riparian system of the area, free from livestock grazing. Restoration sites were previously degraded by grazing but starting in 2001 livestock were either removed completely from these sites or during the songbird breeding season and completely excluded from major riparian corridors using fencing. These livestock exclusion efforts were an attempt to restore the riparian habitat for chats by allowing the understory shrub layer to regrow. The one currently grazed site experienced grazing before and during the study and was only included in our analysis of the effects of vegetation characteristics on chat reproductive performance.

from 13–25 ha, while one restoration site was 180 ha. The 3 reference sites were 2 provincial parks and a private reserve and have not been subjected to grazing by domestic livestock since being protected in the mid-1950s. The 2 restoration sites contained patches of riparian forest but in the late 1990s had highly degraded shrub layers due to livestock grazing. In 2001, livestock were removed from the larger of the 2 restoration sites during the songbird breeding season (May–August) but were allowed to graze in this site during the nonbreeding season. The smaller restoration site was fenced in 2001 to permanently exclude livestock from the whole site. Fencing was also installed along forested riparian corridors near water (of which most chat territories occurred outside of in adjacent scrub vegetation) at the larger site in 2003 to permanently exclude livestock. These livestock exclusion efforts were an attempt to restore the riparian habitat for chats by allowing the understory shrub layer to regrow (Environment and Climate Change Canada 2016). The currently grazed site contained fragmented patches of riparian forest and shrubs and had been grazed before and during the study. The shrub and herbaceous layers of this site were fragmented and degraded, similar to the restoration sites at the start of the study. Because this site had not been the subject of any restoration activities and we did not monitor it from the start of the study, we only included data from this site in our analysis of the effects of nest site vegetation characteristics on chat reproductive performance (see Analysis). Photographs of the study sites area are available in Forrester et al. (2017).

Field Methods and Study Design

We surveyed 4 of the 6 sites from 2002 to 2014. We were unable to access one reference site after 2007, and could only access the currently grazed site from 2011 to 2013. We visited each site at least once every 4 days to determine the number of breeding chat pairs and which territories were occupied. We defined breeding territories, using the same methods as McKibbin and Bishop (2010), based on georeferenced locations where male chats perched or sang throughout the breeding season. We visited each territory ≥3 times and identified ≥12 perch locations per territory. Male chats are vocal and conspicuous so we are confident that we found all occupied territories at each site in years that they were monitored. Although many chats in our study population were color-banded, we did not include individual identity in any of our analyses because some males were unbanded and female identity at nests was rarely known due to their secretive and skulking nature.

We monitored chat territories and nesting attempts from the beginning of May through mid-August during each year of the study. We spent the most time nest searching in the largest site while dividing our remaining time evenly between the smaller sites. We located nests using behavioral observations of adults and systematic searches following Martin and Geupel's (1993) protocol. We checked nests every 3 days except around estimated fledging dates when we checked them daily to determine the occurrence of brood parasitism by cowbirds, nest success (defined as fledging ≥1 chat), and the number of young fledged (presumed to be the same as the number of nestlings still in the nest on the last nest check before fledging). We considered nests to have failed if they were obviously destroyed or if nestlings were missing >2 days before the anticipated fledging date (day 10; R. McKibbin personal observation). We considered nests successful if they were empty within 2 days of the anticipated fledge date and if fledglings were observed or heard, or if parental behavior indicated the presence of fledglings in the area. To experimentally evaluate the influence of cowbird parasitism on reproductive performance, we removed cowbird eggs or nestlings from some parasitized nests during 2007–2011.

We measured nest site vegetation characteristics in chat breeding territories using standardized methods (Morgan et al. 2006, McKibbin and Bishop 2010). We measured 3 habitat characteristics associated with habitat selection or reproductive performance of chats or other riparian songbirds (Best and Stauffer 1980, Burhans and Thompson III 1999, Ricketts and Ritchison 2000, McKibbin and Bishop 2010, Quinlan and Green 2012). We first estimated the percentage cover of shrubs (all species combined) within 50 m of nests. We used tape measures to determine a 50-m radius around the nest and divided the territory into 4 quadrants. We then visually estimated percent cover (i.e. surface area coverage) within each quadrant and then averaged the percentages from the 4 quadrants to get overall cover estimates for breeding territories. The 50-m radius was chosen to enable comparisons with other studies and because 1.16 ha (~60-m radius) is the mean area that chats in the Okanagan Valley defend as their roughly circular breeding territories (McKibbin and Bishop 2012b). We also estimated the percent cover of all shrubs within 5 m of nests (a commonly used radii for nest patch measurements; Martin and Roper 1988, Dearborn and Sanchez 2001), using the same visual estimation method as for the 50-m shrub measurements. Finally, we determined average nest concealment using the standard method of estimating the percentage of the nest covered by vegetation from 1 m away at nest height from each of the 4 cardinal directions and then averaging those 4 values (Santisteban et al. 2002, Colombelli-Négrel and Kleindorfer 2009). We measured nest site vegetation characteristics at ≥5 nests in every year except 2010; funding and time constraints prevented us from measuring all nests in every year (and measuring any nest site characteristics in 2010). When we could not measure nest site characteristics of all nests in a year, we gave priority to nests in territories that had never been surveyed before because one of the goals of our study was to compare the habitat characteristics of breeding territories that were occupied for the first time earlier in the study (when Yellow-breasted Chat abundance was low) with the habitat characteristics of territories that were occupied for the first time later in the study (when abundance was high).

Analysis

First, as chat abundance increased from 2002 to 2014, we determined if chats preferentially selected breeding territories with certain habitat characteristics relating to shrub cover. For this analysis we used all nests from reference and restoration sites that had nest site vegetation data. Because

TABLE 1. Candidate model sets explaining the year each territory was first occupied by a breeding Yellow-breasted Chat (i.e. TOR) in the Okanagan Valley, British Columbia, Canada, 2002–2014 (n = 134 nests with nest site vegetation characteristics measured in 108 breeding territories). Rest = restoration site or reference site, Shrubs50 = average percent cover of shrubs within a 50-m radius of the nest, Shrubs5 = average percent cover of shrubs within a 5-m radius of the nest, Conceal = average percent nest concealment estimated from 1 m away from the nest in each of the 4 cardinal directions, and * = interaction between terms. Models were ranked using Akaike's information criterion corrected for small sample sizes (AIC_c). K = number of model parameters, LL = model log-likelihood, ΔAIC_c = distance from the top model, and w_i = Akaike weight.

Model	K	LL	ΔAIC_c	w_i
Shrubs50*Rest	5	−339.13	0.00 [a]	0.81
Shrubs50*Rest + Shrubs5*Rest	7	−338.54	3.24	0.16
Shrubs50*Rest + Shrubs5*Rest + Conceal*Rest	9	−338.19	7.10	0.02
Rest	3	−346.74	10.94	0.00
Shrubs50 + Shrubs5 + Conceal + Rest	6	−344.82	13.57	0.00
Shrubs5*Rest	5	−346.38	14.50	0.00
Shrubs50	3	−348.54	14.53	0.00
Conceal*Rest	5	−346.53	14.81	0.00
Shrubs5	3	−349.40	16.25	0.00
Conceal	3	−349.44	16.33	0.00
Shrubs50 + Shrubs5	4	−348.53	16.65	0.00
Shrubs5 + Conceal	4	−349.21	18.01	0.00
Shrubs50 + Shrubs5 + Conceal	5	−348.29	18.32	0.00
Shrubs5*RO + Conceal*RO	7	−346.10	18.35	0.00

[a] Lowest AIC_c = 688.73

we did not monitor the currently grazed site from the first year of our study, we did not know when territories in this site were first occupied so we did not include nests from this site in this analysis. We used generalized linear mixed effects models implemented in package lme4 (Bates et al. 2015) to examine if the 3 nest habitat characteristics described above and habitat restoration predicted the year a breeding territory was first occupied by a chat (i.e. territory occupancy rank [TOR]). Territories with a TOR of 1 were occupied in the first year of the study (2002), while a TOR of 2–13 means the territory was occupied for the first time in the corresponding year from 2003 to 2014. We developed a set of candidate models using different combinations of the nest site vegetation characteristics (continuous variables) and habitat restoration (restoration site/reference site) as fixed effects to explain TOR (Table 1). Because we predicted that habitat improvement in restoration sites would create new suitable breeding territories throughout the study period and therefore weaken the effects of vegetation characteristics on TOR (Figure 1), some models included interaction terms between habitat restoration and nest site characteristics (Table 1). We included study site as a random term in this set of models. Although we measured nest site vegetation at different nests within the same breeding territory a few times (see Results), we did not include breeding territory as a random term because the large number of breeding territories led to model convergence issues. We tested for multicollinearity of nest site characteristics by examining variance inflation factor (VIF; Montgomery et al. 2012) values. VIF values for these variables were well within the tolerance range (1.1–1.4), indicating no issues with multicollinearity (Bowerman and O'Connell 2000).

Using the reduced dataset of nests with corresponding nest site vegetation data, we examined the influence of chat nest site vegetation characteristics on 3 measures of chat reproductive performance: the chance of nest parasitism, daily nest survival, and the number of fledglings from successful nests. This analysis included nests from the currently grazed site because we did not include TOR as an explanatory variable. Using generalized linear mixed effects models, we developed 3 sets of candidate models to explain the 3 measures of reproductive performance using different combinations of the nest site vegetation characteristics (continuous variables; Table 2). We used the logistic-exposure method (Shaffer 2004) to calculate daily nest survival.

Then, using our larger sample of nests where we knew the year that each territory was first occupied, we compared the relative influences of breeding territory selection (i.e. TOR), habitat restoration, and brood parasitism on the 3 measures of chat reproductive performance. Because we did not monitor the currently grazed site from the first year of our study and did not know the TOR for territories in this site, we did not include nests from this site in this analysis. Using generalized linear mixed effects models, we developed 3 sets of candidate models to explain the 3 measures of reproductive performance using different combinations of TOR (continuous variable), habitat restoration (restoration site/reference site), and parasitism (not parasitized/parasitized/parasitized and then cowbird egg(s) or nestling(s) removed; Table 3). We included an interaction between TOR and restoration category (Table 3) in some models because we predicted that habitat improvement in restoration sites would improve habitat and therefore weaken the effects of TOR on reproductive performance in restoration sites (Figure 1). We included study site and year as random terms in these sets of models (Tables 2 and 3).

We conducted all analyses in R (R Core Team 2019). All models were fitted using either a Poisson (TOR, number of fledglings) or binomial (chance of parasitism, daily nest survival) distribution. We used an information-theoretic approach using Akaike's information criterion corrected for small sample sizes (AICc) and Akaike weights (w_i) to evaluate the strength of each model (Burnham and Anderson 2002). We considered parameters as informative based on the weight of evidence (i.e. w_i from models) and if

TABLE 2. Candidate model sets explaining how nest site vegetation characteristics influence 3 measures of the reproductive performance of Yellow-breasted Chats: (**A**) Chance of being parasitized by a Brown-headed Cowbird (n = 144 nests), (**B**) daily nest survival (n = 145 nests), and (**C**) number of fledglings from successful nests (n = 87 nests). Shrubs50 = average percent cover of shrubs within a 50-m radius of the nest, Shrubs5 = average percent cover of shrubs within a 5-m radius of the nest, Conceal = average percent nest concealment estimated from 1 m away from the nest in each of the 4 cardinal directions, and Parasitism = whether a nest was not parasitized, parasitized by a cowbird, or was parasitized and had the cowbird egg/nestling removed. Models were ranked using Akaike's information criterion corrected for small sample sizes (AIC_c). K = number of model parameters, LL = model log-likelihood, ΔAIC_c = distance from the top model, and w_i = Akaike weight.

Model	K	LL	ΔAIC_c	w_i
(A) Chance of nest parasitism				
Null	3	−97.97	0.00 [a]	0.31
Shrubs50	4	−97.19	0.56	0.23
Shrubs5	4	−97.89	1.97	0.11
Conceal	4	−97.92	2.02	0.11
Shrubs50 + Conceal	5	−97.14	2.62	0.08
Shrubs50 + Shrubs5	5	−97.18	2.70	0.08
Shrubs5 + Conceal	5	−97.83	4.00	0.04
Shrubs 50 + shrubs5 + Conceal	6	−97.14	4.80	0.03
(B) Daily nest survival				
Null	3	−121.56	0.00 [b]	0.34
Shrubs5	4	−121.03	1.05	0.20
Conceal	4	−121.55	2.09	0.12
Shrubs50	4	−121.56	2.11	0.12
Shrubs50 + Shrubs5	5	−120.93	2.98	0.08
Shrubs5 + Conceal	5	−121.03	3.18	0.07
Shrubs50 + Conceal	5	−121.55	4.23	0.04
Shrubs50 + Shrubs5 + Conceal	6	−120.93	5.14	0.03
(C) Number of fledglings from successful nests				
Shrubs50	4	−146.07	0.00 [c]	0.28
Null	3	−147.20	0.08	0.27
Shrubs5	4	−146.95	1.77	0.11
Shrubs50 + Conceal	5	−146.01	2.14	0.09
Shrubs50 + Shrubs5	5	−146.06	2.24	0.09
Conceal	4	−147.20	2.26	0.09
Shrubs5 + Conceal	5	−146.95	4.01	0.04
Shrubs50 + Shrubs5 + Conceal	6	−146.01	4.44	0.03

[a] Lowest AIC_c = 202.10.
[b] Lowest AIC_c = 249.28.
[c] Lowest AIC_c = 300.60.

the confidence interval (CI) of the beta parameter estimate (β) did not overlap zero (Arnold 2010). We present model results as β estimates and 95% CIs (Table 4).

RESULTS

Influence of Habitat Characteristics and Habitat Restoration on Breeding Territory Selection

We examined the influence of 3 nest site vegetation characteristics and habitat restoration (i.e. reference vs. restoration sites) on chat breeding TOR (i.e. the year a breeding territory was first occupied) using nest site vegetation data that were collected at 70 nests in 61 breeding territories in reference sites and at 64 nests in 47 territories in restoration sites from 2002 to 2014 (10 ± 8 SD nests per year, range = 0–27). TOR was best explained by the interaction between the average percent cover of shrubs within 50 m of nests and habitat restoration; the most supported model (w_i = 0.81) contained only this significant interaction term (t_{129} = 3.43, P < 0.001; Table 1). In reference sites, territories that had higher shrub cover were occupied by a breeding chat earlier (i.e. had a lower TOR) than territories that had lower shrub cover (t_{67} = −3.75, P < 0.001; Figure 3, Table 4A). We did not observe this pattern in restoration sites, where territories that were occupied for the first time later in the study (i.e. had a higher TOR) did not have lower shrub cover than territories that were occupied earlier (i.e. TOR was not significantly influenced by shrub cover; t_{61} = 0.41, P = 0.69; Figure 3, Table 4A). Models including other nest site vegetation characteristics and their interaction with restoration had little support for explaining TOR (Table 1).

Influence of Nest Site Vegetation Characteristics on Reproductive Performance

We examined the influence of 3 nest site vegetation characteristics on the reproductive performance of chats using 70 nests from 61 breeding territories in reference sites, 64 nests from 47 breeding territories in restoration sites, and 21 nests from 12 breeding territories in the currently grazed site. We measured vegetation at some nests in all years from 2002 to 2014 except 2010 (13 ± 7 SD nests per year excluding 2010, range = 5–27). We found limited evidence that nest site vegetation characteristics influenced the number of fledglings from successful chat nests. The top model for the number of fledglings (w_i = 0.28) only included the percent cover of shrubs (50-m radius) term (Table 2C); shrub cover was positively (but not significantly; t_{86} = 1.51, P = 0.13) correlated with the number of fledglings (Figure 4, Table 4B).

We did not find strong evidence that nest site vegetation characteristics influenced the chance of nest parasitism or daily nest survival, as the null models were the top models for both of these measures of reproductive performance (Tables 2A, 2B and 4B). For the chance of nest parasitism, the model with only the percent cover of shrubs (50-m radius) term had some support (w_i = 0.23, Table 2A); parasitized nests had slightly (but not significantly; t_{140} = −1.22, P = 0.22) lower shrub cover than unparasitized nests (43.0 ± 14.2% vs. 45.1 ± 16.3%, respectively; Table 4B). Additionally, for daily nest survival, the model with only the percent cover of shrubs (5-m radius) term had some support (w_i = 0.20, Table 2B); shrub cover (5-m radius) was negatively (but not significantly;

TABLE 3. Candidate model sets explaining how TOR, habitat restoration, and brood parasitism influence 3 measures of the reproductive performance of Yellow-breasted Chats: (**A**) chance of being parasitized by a Brown-headed Cowbird (n = 464, 151 territories), (**B**) daily nest survival (n = 451 nests, 133 territories), and (**C**) number of fledglings from successful nests (n = 270 nests, 113 territories). TOR = territory occupancy rank (i.e. the year that a territory was occupied by a breeding Yellow-breasted Chat for the first time during 2002–2014), Rest = restoration site or reference site, TOR*Rest = interaction between those fixed effects, and Parasitism = whether a nest was not parasitized, parasitized by a cowbird, or was parasitized and had the cowbird egg/nestling removed. Models were ranked using Akaike's information criterion corrected for small sample sizes (AIC$_c$). K = number of model parameters, LL = model log-likelihood, ΔAIC$_c$ = distance from the top model, and w_i = Akaike weight.

Model	K	LL	ΔAIC$_c$	w_i
(A) Chance of nest parasitism				
Null	3	−316.66	0.00 [a]	0.44
TOR	4	−316.15	1.02	0.26
Rest	4	−316.66	2.03	0.16
TOR + Rest	5	−316.15	3.06	0.10
TOR*Rest	6	−316.00	4.81	0.04
(B) Daily nest survival				
Rest + Parasitism	6	−331.97	0.00 [b]	0.41
Parasitism	5	−331.44	0.89	0.26
TOR + Rest + Parasitism	7	−331.96	2.04	0.15
TOR + Parasitism	6	−333.40	2.86	0.10
TOR*Rest + Parasitism	8	−331.62	3.44	0.07
Rest	4	−338.63	9.23	0.00
Null	3	−339.88	9.69	0.00
TOR + Rest	5	−338.59	11.20	0.00
TOR	4	−339.88	11.72	0.00
TOR*Rest	6	−338.06	12.19	0.00
(C) Number of fledglings from successful nests				
Parasitism	5	−429.31	0.00 [c]	0.47
Rest + Parasitism	6	−429.04	1.54	0.22
TOR + Parasitism	6	−429.29	2.05	0.17
TOR + Rest + Parasitism	7	−429.03	3.64	0.08
TOR*Rest + Parasitism	8	−428.03	3.75	0.07
Null	3	−437.63	12.49	0.00

[a] Lowest AIC$_c$ = 639.37.
[b] Lowest AIC$_c$ = 676.13.
[c] Lowest AIC$_c$ = 868.85.

t_{149} = −1.02, P = 0.31) correlated with daily nest survival (Table 4B).

Influence of Breeding Territory Selection, Habitat Restoration, and Brood Parasitism on Reproductive Performance

We examined the influence of chat breeding territory selection (i.e. TOR), habitat restoration, and brood parasitism by cowbirds on the reproductive performance of chats using 259 nests from 84 breeding territories in reference sites and 205 nests from 61 breeding territories in restoration sites (36 ± 20 SD nests per year, range = 5–71). During 2002–2014, 49% (n = 414) of nests of this population were parasitized by cowbirds. We removed cowbird egg(s) or nestling(s) from 63 of those nests during 2007–2011. We found no evidence that the chance of nest parasitism significantly differed between reference and restoration sites (Figure 5A, Tables 3A and 4C). The model that included only the TOR term had some support for an influence on the chance of nest parasitism (ΔAIC$_c$ = 1.02, w_i = 0.26; Table 3A); territories that were occupied for the first time later in the study had a slightly (but not significantly; t_{460} = 1.02, P = 0.31) higher chance of parasitism (Figure 5A, Table 4C). However, the null model had more support (Table 3A), indicating that TOR and habitat restoration did not strongly influence the chance of nest parasitism.

Over the study period, mean daily nest survival was 0.974 (0.969–0.977 95% CI, n = 451). We found evidence that habitat restoration and brood parasitism influenced daily nest survival; the top model (w_i = 0.41) for daily nest survival contained the habitat restoration and the parasitism terms (Table 3B). Daily nest survival was significantly lower in restoration sites (mean 0.963, 0.954–0.971 95% CI, n = 199) than in reference sites (mean 0.977, 0.971–0.982 95% CI, n = 252; t_{445} = −2.51, P = 0.01; Figure 5B, Table 4C). We also found that parasitized nests had lower daily nest survival (mean 0.963, 0.952–0.971 95% CI, n = 159) than unparasitized nests (mean 0.973, 0.966–0.978 95% CI, n = 234; t_{445} = −2.04, P = 0.04; Figure 6A, Table 4C) and that removing cowbird egg(s) or nestling(s) improved daily nest survival (mean 0.984, 0.972–0.991 95% CI, n = 58; t_{445} = 2.31, P = 0.02; Figure 6A, Table 4C). We did not find evidence that chats breeding in territories that were occupied earlier in the study had higher daily nest survival than those breeding in territories that were selected later (no model with TOR term had ΔAIC$_c$ < 2; Table 3B).

Finally, we found evidence that brood parasitism was the only strong influence on the number of fledglings from successful nests. Successful nests fledged an average of 2.72 ± 0.06 SE (n = 270) chat fledglings, unparasitized nests fledged 3.10 ± 0.08 (n = 143) chat fledglings, parasitized nests fledged 2.32 ± 0.10 (n = 87) chat fledglings, and parasitized nests from which we removed cowbird egg(s) or nestling(s) fledged 2.23 ± 0.14 (n = 40) chat fledglings (Figure 6B). Both parasitized nests (t_{265} = −3.42, P < 0.001) and parasitized nests that we removed cowbird egg(s) or nestling(s) from (t_{265} = -2.87, P < 0.01) fledged significantly fewer fledglings than unparasitized nests (Figure 6B, Table 4C). We found weak evidence that successful chat nests in restoration sites fledged fewer fledglings (2.67 ± 0.09 standard error [SE], n = 104) than in reference sites (2.75 ± 0.08 SE, n = 166; Table 4C); the second most supported model contained the parasitism and the habitat restoration terms (Table 3C). However, the effect size of this restoration term was small and had a CI that overlapped zero (Table 4C), indicating that site category did not strongly influence the number of fledglings. TOR also did not strongly influence the number of fledglings

TABLE 4. Parameter estimates from the generalized linear mixed effects models for the 3 separate analyses: (**A**) the effect of nest site vegetation characteristics and habitat restoration on TOR (i.e. the year that a territory was occupied by a breeding Yellow-breasted Chat for the first time during 2002–2014), (**B**) the effect of nest site vegetation characteristics on chat reproductive performance, and (**C**) the effect of habitat restoration, TOR, brood parasitism, and removal of a cowbird egg/nestling from a parasitized nest on chat reproductive performance. Shrubs50 = average percent cover of shrubs within a 50-m radius of the nest, Shrubs5 = average percent cover of shrubs within a 5-m radius of the nest, Conceal = average percent nest concealment estimated from 1 m away from the nest in each of the 4 cardinal directions, Parasitism = brood parasitism by a Brown-headed Cowbird, Cowbird removal = the removal of a cowbird egg/nestling removed from a parasitized chat nest, Rest = restoration site or reference site, and * = interaction between those fixed effects. Estimates are from the top-ranked model for each candidate set (Tables 1–3) that each fixed effect was included in. Values in bold indicate a 95% CI that does not overlap zero.

	Fixed effect	Estimate (β)	SE	95% CI
(A) Influence of habitat characteristics and habitat restoration on TOR				
TOR (all sites)	Shrubs50*Rest [a]	0.021	0.006	**{0.009, 0.033}**
Reference sites	Shrubs50	–0.018	0.005	**{–0.028, –0.008}**
Restoration sites	Shrubs50	0.001	0.003	{–0.005, 0.007}
(B) Influences of nest site vegetation characteristics on chat reproductive performance				
Chance of nest parasitism	Shrubs50	–0.016	0.013	{–0.041, 0.010}
	Shrubs5	–0.003	0.008	{–0.019, 0.013}
	Conceal	–0.002	0.008	{–0.018, 0.013}
Daily nest survival	Shrubs50	0.001	0.010	{–0.019, 0.020}
	Shrubs5	–0.007	0.007	{–0.022, 0.007}
	Conceal	0.001	0.006	{–0.011, 0.013}
Number of fledglings from successful nests	Shrubs50	0.006	0.004	{–0.002, 0.014}
	Shrubs5	0.002	0.003	{–0.004, 0.008}
	Conceal	–0.001	0.003	{–0.006, 0.005}
(C) Influences of habitat restoration, TOR, and brood parasitism on chat reproductive performance				
Chance of nest parasitism	Rest	–0.021	0.276	{–0.562, 0.520}
	TOR	0.041	0.04	{–0.037, 0.119}
	Rest*TOR	–0.041	0.074	{–0.186, 0.104}
Daily nest survival	Rest	–0.455	0.181	**{–0.810, –0.100}**
	TOR	–0.006	0.035	{–0.075, 0.063}
	Rest*TOR	–0.059	0.072	{–0.200, 0.082}
	Parasitism [b]	–0.362	0.177	**{–0.709, –0.015}**
	Cowbird removal [c]	0.796	0.345	**{0.120, 1.472}**
Number of fledglings from successful nests	Rest	–0.057	0.076	{–0.206, 0.092}
	TOR	–0.003	0.014	{–0.030, 0.024}
	Rest*TOR	0.04	0.028	{–0.015, 0.095}
	Parasitism	–0.291	0.085	**{–0.458, –0.124}**
	Cowbird removal	–0.333	0.116	**{–0.560, –0.106}**

[a] Reference level for "Rest" (reference vs. restoration site) set for reference sites.
[b] Reference level for "Parasitism" set to unparasitized nests.
[c] Reference level for "Cowbird removal" set to unparasitized nests.

from successful nests, although the number of fledglings was slightly positively correlated with TOR in restoration sites and slightly negatively correlated with TOR in reference sites (Figure 5C). However, the model that included the interaction between TOR and restoration category was not well supported ($\Delta AIC_c > 2$; Table 3C), so we did not find evidence that the effect of TOR on the number of fledglings differed between reference and restoration sites.

DISCUSSION

The endangered chat population in the south Okanagan Valley has been the subject of conservation actions that have resulted in significant increases in abundance, from near extirpation, since the early 2000s (Forrester et al. 2017). Due to the fragmented nature of the remnant riparian vegetation of the Okanagan Valley (Lea 2008), as well as the high rates of brood parasitism on riparian songbirds in this area (Ward and Smith 2000, Forrester et al. 2017), it was unclear if this rapidly increasing population would be able to successfully breed in this habitat. Degraded habitat within a developed urban matrix, accompanied by high rates of brood parasitism, may function as an ecological trap that can only be supported by immigration (Schlaepfer et al. 2002). Here, we used the unique opportunity offered by the rapid increase in chat abundance that occurred during our study to evaluate the influences of habitat selection, habitat restoration, and brood parasitism on the reproductive performance of chats in this fragmented riparian vegetation. We found that chats preferred to settle in territories with high percent shrub cover,

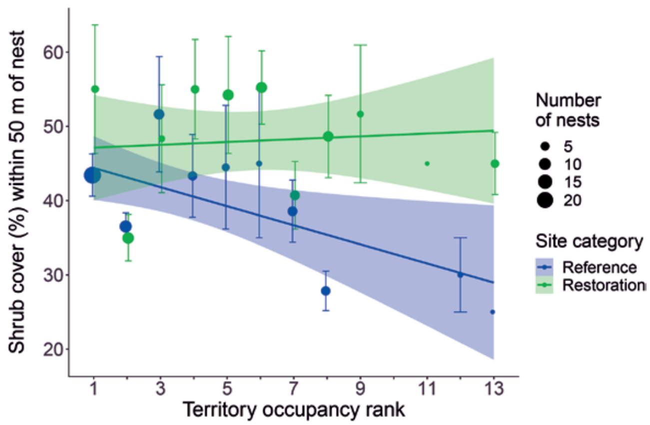

FIGURE 3. Percent shrub cover within 50 m of chat nests in relation to TOR (i.e. the year a territory was first occupied) in reference and restoration sites in the Okanagan Valley. Territories with a TOR of 1 were occupied for the first time in the first year of the study (2002) or earlier, while a TOR of 2–13 means the territory was occupied for the first time in the corresponding year during 2003–2014. Dots show mean values ± SE while the trendline and shaded areas show predicted values and 95% CIs.

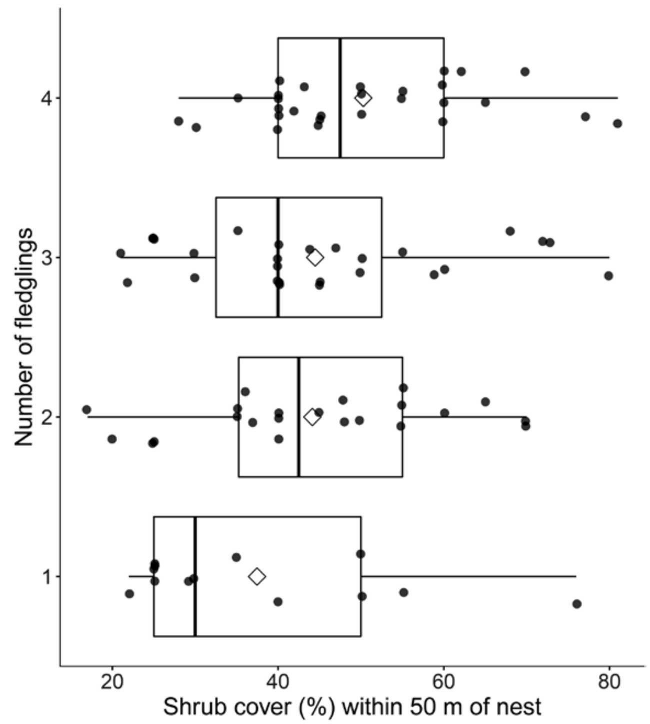

FIGURE 4. Number of fledglings from successful Yellow-breasted Chat nests in relation to shrub cover within 50 m of nests in all study sites, 2002–2014. Boxes represent the interquartile range, thick vertical lines within boxes are the median, diamonds are the mean, and dots are raw data.

and restoration activities enabled chats to continually select breeding territories with high percent shrub cover. Yet, we found that the persistent high rate of brood parasitism by cowbirds, and not habitat selection or habitat restoration, was the dominant influence on chat reproductive performance. Despite this, chat reproductive performance was generally high throughout our 13-yr study period.

We found that chats preferentially selected territories with high percent shrub cover in reference sites, because territories that were first occupied by a breeding chat earlier in the study had higher shrub cover than territories that were first occupied later in the study. It appears that in reference sites where restoration did not take place, established territory holders excluded newly arriving chats from high-quality territories with high shrub cover, consistent with the ideal despotic distribution theory (Fretwell and Lucas 1969, Ferrer and Donazar 1996, Petit and Petit 1996). However, we also found that chats were continually able to select territories with high shrub cover in restoration sites, even as abundance rapidly increased over 13 yr. These results support our prediction that as abundance increased, newer territories would have less preferred habitat characteristics than older territories in reference sites but not in restoration sites. Other studies in western riparian areas also found that removal of livestock resulted in increased shrub cover (Schulz and Leininger 1990, Earnst et al. 2012) and the importance of shrub cover on a territory scale for chats in the Okanagan Valley is in agreement with studies of other western riparian songbirds (Knopf and Sedgwick 1992, Sanders and Edge 1998, Quinlan and Green 2012, Rockwell and Stephens 2017). The increasing availability of areas with high shrub cover as a result of restoration activities is therefore a likely explanation for why chat abundance increased so markedly in restoration sites but not in reference sites (Forrester et al. 2017). Since 10% of nestling chats in this population returned to their natal study site to breed (McKibbin and Bishop 2012a), this rapid increase in chat abundance was likely driven by both natal philopatry and newly colonizing adults.

We did not find strong evidence that nest site vegetation characteristics influenced the reproductive performance of chats, which was counter to our predictions. We did find some evidence that shrub cover (within a 50-m radius of nests) was positively correlated with the number of fledglings from successful nests, but this relationship was not significant. However, we did not find strong evidence that nest site vegetation characteristics influenced the chance of nest parasitism or daily nest survival. This was counter to other studies of Yellow-breasted Chats. In Missouri, the chance of brood parasitism and nest success were both positively correlated with the area of the patch of shrubs that nests were located in (Burhans and Thompson III 1999). In Kentucky, nest success was negatively correlated with shrub cover within 11.3 m of the nest and positively correlated with nest height (Ricketts and Ritchison 2000). However, shrub cover positively influenced the number of fledglings of Yellow Warblers (*Setophaga petechia*) in

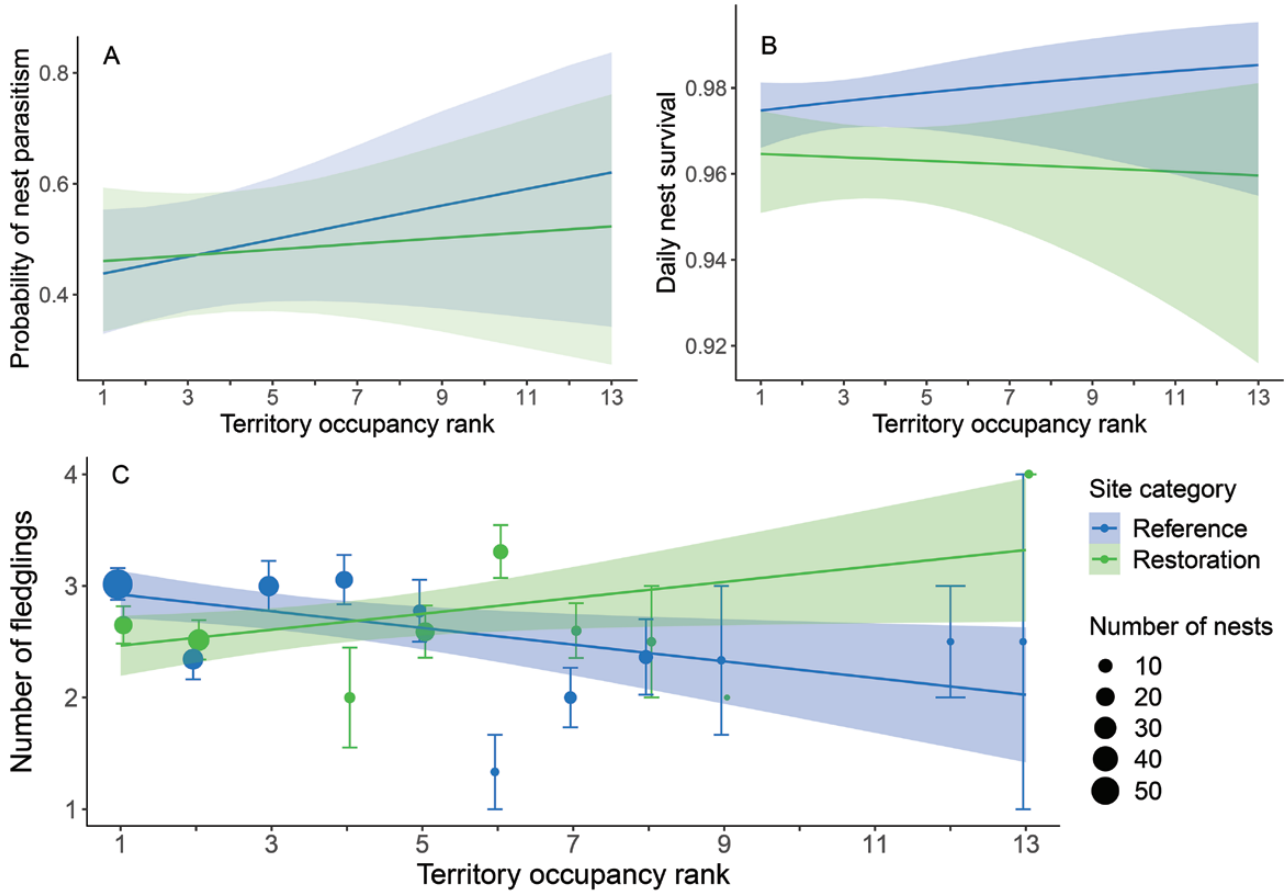

FIGURE 5. (**A**) The chance of nest parasitism by Brown-headed Cowbirds, (**B**) daily nest survival, and (**C**) number of fledglings from successful nests of Yellow-breasted Chats, 2002–2014, in relation to TOR (i.e. the year a territory was first occupied) and site category (reference vs. restoration). Territories with a TOR of 1 were occupied for the first time in the first year of the study (2002) or earlier, while a TOR of 2–13 means the territory was occupied for the first time in the corresponding year during 2003–2014. Dots show mean values ± SE while the trendline and shaded areas show predicted values and 95% CIs.

interior British Columbia (Quinlan and Green 2012). This emphasizes the importance of high percent shrub cover for western riparian songbirds, which often breed in highly fragmented and urbanized riparian corridors where large tracts of shrub cover are rare. Our results also suggest that songbird species that occupy large ranges may have different relationships between their reproductive success and their habitat in different areas of their range.

In contrast to the many studies that have examined the influence of territory occupancy patterns within years on songbird reproductive performance (e.g., Møller 1994, Currie et al. 2000), we examined territory occupancy in terms of the year that territories were first occupied by a breeding chat as abundance increased. Increased chat abundance could have led to a decrease in reproductive performance of the population because of increased competition or because newly arriving males could have been forced into lower quality territories by established males (Ferrer and Donazar 1996, Krüger et al. 2012), as has been demonstrated in other birds (Arcese et al. 1992, Petit and Petit 1996, Both 1998). The lower percent shrub cover of territories that were settled later in the study in reference sites could have resulted in lower reproductive performance, which would be expected under the ideal despotic distribution theory if chat habitat preferences were correlated with reproductive performance. However, we found only limited evidence that breeding territory selection by chats was correlated with reproductive performance. This may have been because the generally disturbed and fragmented riparian landscape and, despite restoration, limited nesting sites resulted in chats being unable to assess or select high-quality breeding territories. This can result when habitat disturbance alters the evolutionary relationships between predators and habitat structure cues that animals use to make settlement decisions (Misenhelter and Rotenberry 2000, Barea

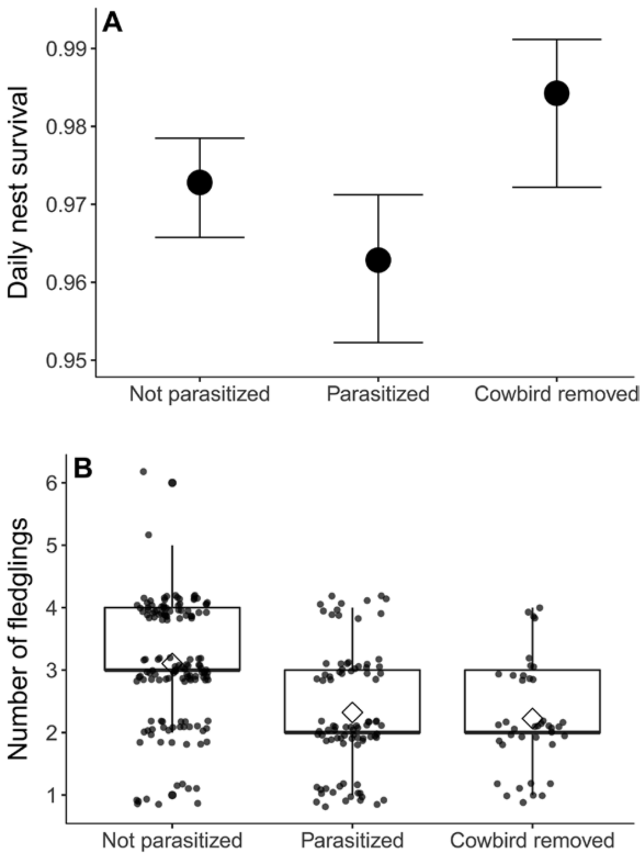

FIGURE 6. (**A**) Daily nest survival (error bars illustrate 95% CIs; n = 234 not parasitized, 159 parasitized, and 58 cowbird removed nests) of Yellow-breasted Chats and (**B**) number of fledglings (boxes represent the interquartile range, thick vertical lines within boxes are the median, diamonds are the mean, and dots are raw data; n = 143 not parasitized, 87 parasitized, and 40 cowbird removed nests) from successful Yellow-breasted Chat nests that were not parasitized, were parasitized by a Brown-headed Cowbird, and for nests that had cowbird egg(s) or nestling(s) removed.

and Watson 2013). Indeed, persistent high rates of brood parasitism in our study sites may have contributed to this disconnect between habitat characteristics and chat reproductive performance.

Despite the fact that chat abundance increased far more in restoration sites than in reference sites (Forrester et al. 2017), daily nest survival was actually lower in restoration sites. This may have been because of an increased number of inexperienced second-year chats (hatched the previous year) returning to restoration sites (McKibbin and Bishop 2012b) or a more intact predator community in the restoration sites. Although restoration efforts did not appear to improve the success or productivity of individual nesting attempts, it did enable chats to breed in areas that were previously unsuitable, so restoration appeared to increase the overall productivity of this endangered population. As our results come from only 2 restoration sites (of which one is much larger than the other sites), future Yellow-breasted Chat management in the Okanagan Valley could benefit from the establishment of additional restoration sites and the monitoring of habitat selection and reproductive performance in riparian sites of a greater variety of sizes. Additionally, post-fledgling survival has a large impact on population growth rates (Clark and Martin 2007, Mattsson and Cooper 2007), and habitat characteristics, even at the landscape scale, can influence post-fledgling survival (Mitchell et al. 2010, Vernasco et al. 2018). Fledglings may use different habitat than breeding adults (King et al. 2006, Streby et al. 2014, Raybuck et al. 2020), but the influences of habitat characteristics and habitat restoration on post-fledging survival are poorly understood for riparian songbirds in general (Vormwald et al. 2011) and unknown for Yellow-breasted Chats. Thus, studies of the relationship between habitat characteristics and survival in this critical post-fledging life-stage, especially in fragmented western landscapes where riparian vegetation is often disturbed and limited, will also benefit conservation efforts for Yellow-breasted Chats and other riparian songbirds.

We found that nest parasitism by cowbirds, which commonly occurs to riparian songbirds in our study sites (Ward and Smith 2000, Forrester et al. 2017), was the main factor influencing chat reproductive performance. Our best supported models indicated that parasitized nests had lower daily nest survival and fewer fledglings than unparasitized nests, likely because cowbird nestlings often outcompete host nestlings and cowbird females usually remove at least one host egg when laying their own (Weatherhead 1989, Lorenzana and Sealy 1999). However, this reduction in the number of chat fledglings was not as severe as that reported for other parasitized songbirds (Lorenzana and Sealy 1999, Benson et al. 2010). Our analysis also demonstrated that experimentally removing cowbird eggs or nestlings from parasitized nests improved daily nest survival, although the best supported model did not include this term, which adds uncertainty to this conclusion. However, we did not find any evidence that removing cowbird eggs or nestlings from parasitized nests increased the number of chat fledglings. These observational and experimental results therefore indicate that the primary negative influence of parasitism on chat reproductive performance in our study population is through the reduction in clutch size that results when cowbirds remove chat eggs to lay their own. However, parasitism may have negatively affected other aspects of reproductive performance that we did not measure, such as nestling growth rates and fledgling survival (Smith 1981, Marvil and Cruz 1989, Burhans et al. 2000). Although cowbirds have likely occurred in the Okanagan Valley long enough for host species to have evolved defenses (Ward and Smith 1998) and cowbirds have decreased in abundance over time in the Okanagan

Valley (Morgan et al. 2006, Forrester et al. 2017), the riparian habitat may be too small and fragmented for chats to be able to select habitat that reduces the chance of parasitism. However, persistent parasitism did not prevent this chat population in the south Okanagan Valley from increasing dramatically over the study period. While productivity was depressed by cowbirds, there appears to have been enough suitable territories and successful nests to consistently increase recruitment into the breeding population during 2002–2014.

Despite the fact that habitat selection or habitat restoration did not appear to strongly influence chat reproductive performance, chats in this population are still relatively successful compared with other populations of chats and other riparian songbirds. Daily nest survival of chats in the Okanagan (0.974) was higher when compared with estimates from other locations calculated using the logistic-exposure or the Mayfield method (range: 0.94–0.96; Burhans and Thompson III 1999, Ricketts and Ritchison 2000, Whitehead et al. 2000, Roach et al. 2018, Stephens and Rockwell 2019), and with estimates of apparent nesting success from other locations (Thompson and Nolan, Jr. 1973, Tewksbury et al. 2006, Forsman and Martin 2009, Stumpf et al. 2012). Chats in the Okanagan also produced a similar or higher number of fledglings per successful nest (overall mean: 2.72; unparasitized nests: 3.10; parasitized nests: 2.32) compared with other studies (unparasitized range: 1.4–2.9, parasitized range: 1.4–2.3; Averill 1996, Burhans and Thompson III 1999, Whitehead et al. 2000). Rates of brood parasitism of chats from other locations range widely (0–80%; Brown 1994, Averill 1996, Burhans and Thompson III 1999, Ricketts and Ritchison 2000, Whitehead et al. 2000), so the 49% rate of parasitism of our population is high but not abnormal. Nest success and productivity of chats in the Okanagan Valley were also generally high compared with other western songbirds (Brown 1994, Averill 1996, Powell and Steidl 2000, Tewksbury et al. 2006, Stumpf et al. 2012). Although this comparison is not a formal test to examine if the fragmented riparian vegetation of the Okanagan Valley is an ecological trap because we did not measure population growth rates, this study provides evidence that chats can breed successfully and have high productivity in such fragmented landscapes if suitable habitat is available.

In conclusion, as the population of chats in the south Okanagan Valley increased rapidly from 2002 to 2014, they preferentially selected breeding territories with high shrub cover. Yet, our analyses of the influences of nest site vegetation characteristics, breeding territory selection, habitat restoration, and brood parasitism did not provide strong evidence that habitat selection and restoration influenced chat reproductive performance. Instead, the high rate of brood parasitism was the dominant influence on the reproductive performance of chats in this population. Regardless, restoration efforts appeared to create additional chat breeding habitat and the high rate of nest parasitism did not prevent this population from increasing rapidly over the study period. This endangered population also had relatively high nest success and productivity compared with other chat populations. This study therefore demonstrates that Yellow-breasted Chats can breed successfully in small patches of riparian habitat within a heavily developed and fragmented landscape. This suggests that conservation efforts that restore or conserve the shrub layer of riparian areas, even in small habitat fragments embedded within heavily developed areas, can be beneficial to shrub-nesting riparian bird populations, but there is a need to quantify how strongly cowbird parasitism affects the reproductive performance of target species.

ACKNOWLEDGMENTS

We thank C. Aikins, R. Armstrong, D. Brown, O. Dyer, J. Emery, R. Hall, A. C. Huang, M. Lancaster, H. Lefort, T. C. Morgan, J. Ruddy, J. St-Amand, and many others for support and assistance with data collection. We also thank the En'owkin Centre, the Penticton Indian Band, the Osoyoos Indian Band, Environment and Climate Change Canada, the Nature Trust of British Columbia, the South Okanagan-Similkameen Conservation Program, Ducks Unlimited, the Nature Conservancy of Canada, the Ministry of Forests, Lands and Natural Resources Operations, Partners in Flight BC/Yukon, BC Parks, the BC Ministry of Water, Land and Air Protection, the Federation of BC Naturalists, the BC Important Bird Areas Program, the Canadian Nature Federation, Bird Studies Canada, Birdlife International, and the BC Conservation Foundation for additional support. We also thank the private landowners who provided access to their properties and participated in riparian restoration efforts. This work benefited from early conversations with K. Martin and R. C. Ydenberg, while comments from J. C. Mouton, A. C. Doll, and an anonymous reviewer improved the manuscript. The authors of this paper declare no conflict of interest.

Funding statement: This study was funded by the Federal Department of Environment and Climate Change Canada (Grant #1100451), the Natural Sciences and Engineering Research Council of Canada (Grant #RGPIN 261899), Simon Fraser University, the Centre for Wildlife Ecology, and an NSERC Discovery grant (Grant #261899). Funders had no input into the content of the manuscript and did not require their approval of the manuscript before submission or publication.

Ethics statement: The Simon Fraser University Animal Care Committee (permit #1039B-12) and Environment and Climate Change Canada (Federal Species at Risk Act-compliant migratory bird permits #59-01- 0433, 59-03-0426, 59-04-0362, 59-05-0378, 59-06-0304, 59- 07-0279, BC-09-0307, BC-09-0144, BC-10–0050, BC-13–0028, BC-14–0034) approved all field methods.

Author contributions: T.R.F., D.J.G, and C.A.B. conceived the idea, design, and experiment (supervised research, formulated questions and hypotheses); T.R.F., R.M., A.M.B., and C.A.B. performed the experiments (collected data, conducted the research); T.R.F., D.J.G., R.M., A.M.B., and C.A.B. wrote (or substantially edited) the paper; T.R.F., D.J.G., R.M., and C.A.B. developed or designed methods; T.R.F., D.J.G., and C.A.B. analyzed the data; D.J.G., R.M., A.M.B., and C.A.B. contributed substantial materials, resources, or funding.

Data Availability: Analyses reported in this article can be reproduced using the data provided by Forrester et al. (2020).

LITERATURE CITED

Andrén, H. (1990). Despotic distribution, unequal reproductive success, and population regulation in the Jay *Garrulus Glandarius L*. Ecology 71:1796–1803.

Arcese, P., J. N. M. Smith, W. M. Hochachka, C. M. Rogers, and D. Ludwig (1992). Stability, regulation, and the determination of abundance in an insular Song Sparrow population. Ecology 73:805–822.

Arnold, T. W. (2010). Uninformative parameters and model selection using Akaike's Information Criterion. Journal of Wildlife Management 74:1175–1178.

Averill, A. (1996). Brown-headed Cowbird parasitism of neotropical migratory songbirds in riparian areas along the lower Colorado river. M.S. Thesis, The University of Arizona, Tucson, AZ, USA.

Barea, L. P., and D. M. Watson (2013). Trapped between popular fruit and preferred nest location—cafeterias are poor places to raise a family. Functional Ecology 27:766–774.

Bates, D., M. Mächler, B. M. Bolker, and S. C. Walker (2015). Fitting linear mixed-effects models using lme4. Journal of Statistical Software 67:1–48.

Battin, J. (2004). When good animals love bad habitats: Ecological traps and the conservation of animal populations. Conservation Biology 18:1482–1491.

Benson, T. J., N. M. Anich, J. D. Brown, and J. C. Bednarz (2010). Habitat and landscape effects on brood parasitism, nest survival, and fledgling production in Swainson's Warblers. Journal of Wildlife Management 74:81–93.

Best, L. B., and D. F. Stauffer (1980). Factors affecting nesting success in riparian bird communities. The Condor 82:149–158.

Bock, C. E., and Z. F. Jones (2004). Avian habitat evaluation: Should counting birds count? Frontiers in Ecology and the Environment 2:403–410.

Both, C. (1998). Density dependence of clutch size: Habitat heterogeneity or individual adjustment? Journal of Animal Ecology 67:659–666.

Bowerman, B. L., and R. T. O'Connell (2000). Linear Statistical Models: An Applied Approach. 2nd edn. Duxbury Press, Belmont, CA, USA.

Brown, B. T. (1994). Rates of brood parasitism by Brown-headed Cowbirds on riparian passerines in Arizona. Journal of Field Ornithology 65:160–168.

Burhans, D. E., and F. R. Thompson III (1999). Habitat patch size and nesting success of Yellow-breasted Chats. The Wilson Bulletin 111:210–215.

Burhans, D. E., F. R. Thompson III, and J. Faaborg (2000). Costs of parasitism incurred by two songbird species and their quality as cowbird hosts. The Condor 102:364–373.

Burnham, K. P., and D. R. Anderson (2002). Model Selection and Multimodel Inference: A Practical Information-Theoretic approach. 2nd edn. Springer-Verlag, New York, NY, USA.

Cannings, R. A., R. J. Cannings, and S. G. Cannings (1987). Birds of the Okanagan Valley, British Columbia. University of British Columbia Press, Vancouver, Canada.

Cannings, R. J., C. Charlesworth, and C. Siddle (2010). Checklist of the Birds of the Okanagan Valley. South Okanagan Naturalists Club, Penticton, Canada.

Chalfoun, A. D., and K. A. Schmidt (2012). Adaptive breeding-habitat selection: Is it for the birds? The Auk 129:589–599.

Clark, M. E., and T. E. Martin (2007). Modeling tradeoffs in avian life history traits and consequences for population growth. Ecological Modelling 209:110–120.

Clark, R. G., and D. Shutler (1999). Avian habitat selection: Pattern from process in nest-site use by ducks? Ecology 80:272–287.

Colombelli-Négrel, D., and S. Kleindorfer (2009). Nest height, nest concealment, and predator type predict nest predation in Superb Fairy-wrens (*Malurus cyaneus*). Ecological Research 24:921–928.

Currie, D., D. B. A. Thompson, and T. Burke (2000). Patterns of territory settlement and consequences for breeding success in the Northern Wheatear *Oenanthe oenanthe*. Ibis 142:389–398.

Davis, J. B., M. Guillemain, M. Richard, C. Arzel, J. M. Eadie, and E. C. Rees (2014). Habitat and resource use by waterfowl in the northern hemisphere in autumn and winter. Wildfowl 4:17–69.

Dearborn, D. C., and L. L. Sanchez (2001). Do Golden-cheeked Warblers select nest locations on the basis of patch vegetation? The Auk 118:1052–1057.

Donovan, T. M., F. R. Thompson III, J. Faaborg, J. R. Probst, and J. R. Probstt (1995). Reproductive success of migratory birds in habitat sources and sinks. Conservation Biology 9:1380–1395.

Earnst, S. L., D. S. Dobkin, and J. A. Ballard (2012). Changes in avian and plant communities of aspen woodlands over 12 years after livestock removal in the northwestern Great Basin. Conservation Biology 26:862–872.

Environment and Climate Change Canada (2016). Recovery Strategy for the Yellow-breasted Chat *auricollis* subspecies (*Icteria virens auricollis*) (Southern Mountain population) in Canada. Species at Risk Act Recovery Strategy Series. Government of Canada, Ottawa.

Ferrer, M., and J. A. Donazar (1996). Density-dependent fecundity by habitat heterogeneity in an increasing population of Spanish Imperial Eagles. Ecology 77:69–74.

Forrester, T. R., D. J. Green, R. McKibbin, A. M. Bezener, and C. A. Bishop (2020). Data from: Riparian habitat restoration increases the availability and occupancy of Yellow-breasted Chat territories but brood parasitism is the primary influence on reproductive performance. The Condor: Ornithological Applications 122:1–16. doi: 10.5061/dryad.prr4xgxj9.

Forrester, T. R., D. J. Green, R. McKibbin, and C. A. Bishop (2017). Evaluating the efficacy of seasonal grazing and livestock exclusion as restoration tools for birds in riparian habitat of the Okanagan Valley, British Columbia, Canada. Restoration Ecology 25:768–777.

Forsman, J. T., and T. E. Martin (2009). Habitat selection for parasite-free space by hosts of parasitic cowbirds. Oikos 118:464–470.

Fretwell, S. D., and H. L. J. Lucas (1969). On territorial behaviour and other factors influencing habitat distribution in birds. I. Theoretical development. Acta Biotheoretica 19:16–36.

Gardali, T., A. L. Holmes, S. L. Small, N. Nur, G. R. Geupel, and G. H. Golet (2006). Abundance patterns of landbirds in restored and remnant riparian forests on the Sacramento River, California, U.S.A. Restoration Ecology 14:391–403.

Germain, R. R., and P. Arcese. (2014). Distinguishing individual quality from habitat preference and quality in a territorial passerine. Ecology 95:436–445.

Hall, L. S., P. R. Krausman, and M. L. Morrison (1997). The habitat concept and a plea for standard terminology. Wildlife Society Bulletin 25:173–182.

Högstedt, G. (1980). Evolution of clutch size in birds: Adaptive variation in relation to territory quality. Science 210:1148–1150.

Johnson, M. D. (2007). Measuring habitat quality: A review. The Condor 109:489–504.

Joos, C. J., F. R. Thompson III, and J. Faaborg (2014). The role of territory settlement, individual quality, and nesting initiation on productivity of Bell's Vireos *Vireo bellii bellii*. Journal of Avian Biology 45:584–590.

King, D. I., R. M. Degraaf, M. L. Smith, and J. P. Buonaccorsi (2006). Habitat selection and habitat-specific survival of fledgling ovenbirds (*Seiurus aurocapilla*). Journal of Zoology 269:414–421.

Knopf, F. L., and J. A. Sedgwick (1992). An experimental study of nest-site selection by Yellow Warblers. The Condor 94:734–742.

Krueper, D., J. Bart, and T. D. Rich (2003). Response of vegetation and breeding birds to the removal of cattle on the San Pedro River, Arizona (U.S.A.). Conservation Biology 17:607–615.

Krüger, O., N. Chakarov, J. T. Nielsen, V. Looft, T. Grünkorn, B. Struwe-Juhl, and A. P. Møller (2012). Population regulation by habitat heterogeneity or individual adjustment? The Journal of Animal Ecology 81:330–340.

Lanyon, S. M., and C. F. Thompson (1986). Site fidelity and habitat quality as determinants of settlement pattern in male painted buntings. The Condor 88:206–210.

Lea, T. (2008). Historical (pre-settlement) ecosystems of the Okanagan Valley and Lower Similkameen Valley of British Columbia—pre-European contact to the present. Davidsonia 19:3–36.

Lorenzana, J. C., and S. G. Sealy (1999). A meta-analysis of the impact of parasitism by the Brown-Headed Cowbird on its hosts. Studies in Avian Biology 18:24–253.

Martin, T. E. (1987). Food as a limit on breeding birds: A life-history perspective. Annual Review of Ecology and Systematics 18:453–487.

Martin, T. E. (1998). Are microhabitat preferences of coexisting species under selection and adaptive? Ecology 79:656–670.

Martin, T. E., and G. R. Geupel (1993). Nest-monitoring plots: Methods for locating nests and monitoring success. Journal of Field Ornithology 64:507–519.

Martin, T. E., and J. J. Roper (1988). Nest predation and nest-site selection of a western population of the Hermit Thrush. The Condor 90:51–57.

Marvil, R. E., and A. Cruz (1989). Impact of Brown-headed Cowbird parasitism on the reproductive success of the Solitary Vireo. The Auk 106:476–480.

Mattsson, B. J., and R. J. Cooper (2007). Which life-history components determine breeding productivity for individual songbirds? A case study of the Louisiana Waterthrush (*Seiurus motacilla*). The Auk 124:1186–1200.

McKibbin, R., and C. A. Bishop (2008). Observations on the longevity and fecundity of the western Yellow-breasted Chat in the south Okanagan Valley, British Columbia, Canada. British Columbia Birds 18:26–27.

McKibbin, R., and C. A. Bishop (2010). Habitat characterization of breeding territories of the Western Yellow-breasted Chat in the South Okanagan, British Columbia, Canada. Northwestern Naturalist 91:145–156.

McKibbin, R., and C. A. Bishop (2012a). Site fidelity and annual survival of the Western Yellow-breasted Chat (*Icteria virens auricollis*) at the northern edge of its range. Canadian Field-Naturalist 126:135–142.

McKibbin, R., and C. A. Bishop (2012b). Size of territories and home ranges of male Western Yellow-breasted Chats (*Icteria virens auricollis*) in British Columbia. The Canadian Field-Naturalist 126:152–156.

Misenhelter, M. D., and J. T. Rotenberry (2000). Choices and consequences of habitat occupancy and nest site selection in Sage Sparrows. Ecology 81:2892–2901.

Mitchell, G. W., P. D. Taylor, and I. G. Warkentin (2010). Multiscale postfledging habitat associations of juvenile songbirds in a managed landscape. The Auk 127:354–363.

Møller, A. P. (1994). Phenotype-dependent arrival time and its consequences in a migratory bird. Behavioral Ecology and Sociobiology 35:115–122.

Montgomery, D. C., E. A. Peck, and G. G. Vining (2012). Introduction to Linear Regression Analysis. 5th edn. Wiley, New York, NY, USA.

Morgan, T. C., C. A. Bishop, and A. M. Bezener (2006). Temporal fluctuation in abundance of Brown-headed Cowbirds and their hosts in riparian habitat in the Okanagan Valley, British Columbia, Canada. Journal of Field Ornithology 77:444–451.

Orians, G. H., and J. F. Wittenberger (1991). Spatial and temporal scales in habitat selection. The American Naturalist 137:S29–S49.

Peak, R. G., and F. R. Thompson III (2014). Seasonal productivity and nest survival of Golden-cheeked Warblers vary with forest type and edge density. The Condor: Ornithological Applications 116:546–559.

Petit, L. J., and D. R. Petit (1996). Factors governing habitat selection by prothonotary warblers: Field tests of the Fretwell-Lucas models. Ecological Monographs 66:367–387.

Powell, B. F., and R. J. Steidl (2000). Nesting habitat and reproductive success of southwestern riparian birds. The Condor 102:823–831.

Quinlan, S. P., and D. J. Green (2012). Riparian habitat disturbed by reservoir management does not function as an ecological trap for the Yellow Warbler (*Setophaga petechia*). Canadian Journal of Zoology 90:320–328.

R Core Team (2019). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. https://www.r-project.org.

Raybuck, D. W., J. L. Larkin, S. H. Stoleson, and T. J. Boves (2020). Radio-tracking reveals insight into survival and dynamic habitat selection of fledgling Cerulean Warblers. The Condor: Ornithological Applications 122:1–15.

Ricketts, M. S., and G. Ritchison (2000). Nesting success of Yellow-Breasted Chats: Effects of nest site and territory vegetation structure. The Wilson Bulletin 112:510–516.

Roach, M. C., F. R. Thompson III, and T. Jones-Farrand (2018). Songbird nest success is positively related to restoration of

pine-oak savanna and woodland in the Ozark Highlands, Missouri, USA. The Condor: Ornithological Applications 120:543–556.
Rockwell, S. M., and J. L. Stephens (2017). Habitat selection of riparian birds at restoration sites along the Trinity River, California. Restoration Ecology 26:767–777.
Sanders, T. A., and W. D. Edge (1998). Breeding bird community composition in relation to riparian vegetation structure in the Western United States. Journal of Wildlife Management 62:461–473.
Santisteban, L., K. E. Sieving, and M. L. Avery (2002). Use of sensory cues by Fish Crows *Corvus ossifragus* preying on artificial bird nests. Journal of Avian Biology 33:245–252.
Schlaepfer, M. A., M. C. Runge, and P. W. Sherman (2002). Ecological and evolutionary traps. Trends in Ecology & Evolution 17:474–480.
Schulz, T. T., and W. C. Leininger (1990). Differences in riparian vegetation structure between grazed areas and exclosures. Journal of Range Management 43:295–299.
Shaffer, T. L. (2004). A unified approach to analyzing nest success. The Auk 121:526–540.
Smith, J. N. M. (1981). Cowbird parasitism, host fitness, and age of the host female in an island Song Sparrow population. The Condor 83:152–161.
Stephens, J. L., and S. M. Rockwell (2019). Short-term riparian restoration success measured by territory density and reproductive success of three songbirds along the Trinity River, California. The Condor: Ornithological Applications 121:1–12.
Streby, H. M., J. M. Refsnider, and D. E. Anderson (2014). Redefining reproductive success in songbirds: Moving beyond the nest success paradigm. The Auk: Ornithological Advances 131:718–726.
Stumpf, K. J., T. C. Theimer, M. A. McLeod, and T. J. Koronkiewicz (2012). Distance from riparian edge reduces brood parasitism of southwestern Willow Flycatchers, whereas parasitism increases nest predation risk. Journal of Wildlife Management 76:269–277.
Tewksbury, J. J., L. Garner, S. Garner, J. D. Lloyd, V. Saab, and T. E. Martin. (2006). Tests of landscape influence: Nest predation and brood parasitism in fragmented ecosystems. Ecology 87:759–768.
Thompson, C. F., and V. Nolan, Jr. (1973). Biology of the Yellow-breasted Chat (*Icteria virens L.*) in southern Indiana. Ecological Monographs 43:145–171.
Tschumi, M., M. Schaub, and R. Arlettaz (2014). Territory occupancy and parental quality as proxies for spatial prioritization of conservation areas. PLoS One 9:1–11.
Van Horne, B. (1983). Density as a misleading indicator of habitat quality. Journal of Wildlife Management 47:893–901.
Vernasco, B. J., T. S. Sillett, P. P. Marra, and T. B. Ryder (2018). Environmental predictors of nestling condition, postfledging movement, and postfledging survival in a migratory songbird, the Wood Thrush (*Hylocichla mustelina*). The Auk: Ornithological Advances 135:15–24.
Vormwald, L. M., M. L. Morrison, H. A. Mathewson, M. Cocimano, and B. A. Collier (2011). Survival and movements of fledgling Willow and Dusky Flycatchers. The Condor 113:834–842.
Ward, D., and J. N. M. Smith (1998). Morphological differentiation of Brown-headed Cowbirds in the Okanagan Valley, British Columbia. The Condor 100:1–7.
Ward, D., and J. N. M. Smith (2000). Interhabitat differences in parasitism frequency by Brown-headed Cowbirds in the Okanagan Valley, British Columbia. In Ecology and Management of Cowbirds and their Hosts (J. N. M. Smith, T. L. Cook, S. I. Rothstein, S. K. Robinson, and S. G. Sealy, Editors). University of Texas Press, Austin, USA. pp. 210–219.
Warman, L. D., D. M. Forsyth, A. R. E. Sinclair, K. Freemark, H. D. Moore, T. W. Barrett, R. L. Pressey, and D. White (2004). Species distributions, surrogacy, and important conservation regions in Canada. Ecology Letters 7:374–379.
Weatherhead, P. J. (1989). Sex ratios, host-specific reproductive success, and impact of Brown-Headed Cowbirds. The Auk 106:358–366.
White, C. F., K. Lyons, S. J. Jorgensen, J. O'Sullivan, C. Winkler, K. C. Weng, and C. G. Lowe. (2019). Quantifying habitat selection and variability in habitat suitability for juvenile white sharks. Plos One 14:e0214642.
Whitehead, M. A., S. H. Schweitzer, and W. Post (2000). Impact of brood parasitism on nest survival parameters and seasonal fecundity of six songbird species in southeastern old-field habitat. The Condor 102:946–950.